

CM0081 Formal Languages and Automata

§ 9.2 An Undecidable Problem That Is Recursively Enumerable

Andrés Sicard-Ramírez

Universidad EAFIT

Semester 2024-1

(Last modification: 12th January 2026)

Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

Theorems About Recursive Languages

Theorem 9.3

If L is a recursive language, then $\overline{L}$ is also a recursive language.

Proof

Whiteboard.

Theorems About Recursive Languages

Theorem 9.3

If L is a recursive language, then $\overline{L}$ is also a recursive language.

Proof

Whiteboard.

Theorem 9.4

If both L and $\overline{L}$ are recursively enumerable languages, then L is recursive (and $\overline{L}$ is recursive as well by Theorem 9.3).

Proof

Whiteboard.

Theorems About Recursive Languages

Possible relations between a language L and its complement $\bar{L}$

- (i) Both L and $\bar{L}$ are recursive.

Theorems About Recursive Languages

Possible relations between a language L and its complement $\overline{L}$

- (i) Both L and $\overline{L}$ are recursive.
- (ii) Neither L nor $\overline{L}$ are recursively enumerable.

Theorems About Recursive Languages

Possible relations between a language L and its complement $\overline{L}$

- (i) Both L and $\overline{L}$ are recursive.
- (ii) Neither L nor $\overline{L}$ are recursively enumerable.
- (iii) L is recursively enumerable but not recursive and $\overline{L}$ is not recursively enumerable.

Theorems About Recursive Languages

Possible relations between a language L and its complement $\overline{L}$

- (i) Both L and $\overline{L}$ are recursive.
- (ii) Neither L nor $\overline{L}$ are recursively enumerable.
- (iii) L is recursively enumerable but not recursive and $\overline{L}$ is not recursively enumerable.
- (iv) $\overline{L}$ is recursively enumerable but not recursive and L is not recursively enumerable.

Theorems About Recursive Languages

Exercise 9.2.5

Let L be recursively enumerable and let $\overline{L}$ be non recursively enumerable. Consider the language

$$L' = \{0w \mid w \text{ is in } L\} \cup \{1w \mid w \text{ is not in } L\}.$$

Can you say for certain whether L' is recursive, recursively enumerable, or non recursively enumerable? Justify your answer.

Solution (from Hopcroft, Motwani and Ullman (2007))

Suppose L' were recursively enumerable. Then we could design a Turing machine M for $\overline{L}$ as follows. Given input w , M changes its input to $1w$ and simulates the hypothetical Turing machine for L' . If that Turing machine accepts, then w is in $\overline{L}$, so M should accept. If the Turing machine for L' never accepts, then neither does M . Thus, M would accept exactly $\overline{L}$, which contradicts the fact that $\overline{L}$ is not recursively enumerable. We conclude that L' is not recursively enumerable.

The Universal Language

Conventions

1. (M, w) : Represents the Turing machine M with input w .
2. w is a string of 0's and 1's.

Codification of a Turing machine with an input

Let w_i be the codification of a Turing machine M . The codification of (M, w) is defined by

$$\overrightarrow{(M, w)} := w_i 111w.$$

The Universal Language

Definition

Let $\Sigma = \{0, 1\}$. The **universal language**, denoted L_u , is the set of binary strings that encode a pair (M, w) such that $w \in L(M)$, that is,

$$L_u := \left\{ \overrightarrow{(M, w)} \in \Sigma^* \mid w \in L(M) \right\}.$$

The Universal Language

Theorem

The language L_u is recursively enumerable.

Idea of the proof

There exists a Turing machine U such that $L_u = L(U)$. The machine U is called a **universal Turing machine**.

The Universal Language

Theorem 9.6

The language L_u is recursively enumerable but not recursive.

Proof of L_u is not recursive (by contradiction (proof of negation))

Suppose L_u is recursive

$\Rightarrow \overline{L_u}$ is recursive

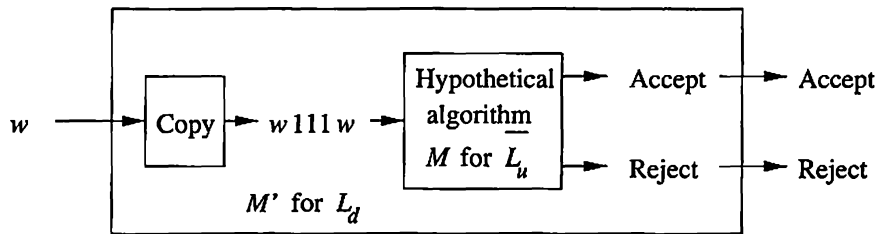
$\Rightarrow L_d$ is recursive (see next slide)

$\Rightarrow$ Contradiction because L_d is non recursively enumerable ■

The Universal Language

From the recursiveness of $\overline{L_u}$ to the recursiveness of L_d

Given a terminating Turing machine for accepting $\overline{L_u}$ we could use this machine for building a terminating Turing machine for accepting L_d (Fig. 9.6).



Since L_d is not recursive (because it is not recursive enumerable) then $\overline{L_u}$ is not recursive.

Code for a Universal Turing Machine

Code for U

Since U is a Turing machine exists i (1654 digits) such that $U = M_i$ given by (using a different codification) (Penrose 1991, pp. 56-57):

```
724485533533931757719839503961571123795236067255655963110814479660650
505940424109031048361363235936564444345838222688327876762655614469281
411771501784255170755408565768975334635694247848859704693472573998858
228382779529468346052106116983594593879188554632644092552550582055598
945189071653741489603309675302043155362503498452983232065158304766414
213070881932971723415105698026273468642992183817215733348282307345371
342147505974034518437235959309064002432107734217885149276079759763441
512307958639635449226915947965461471134570014504816733756217257346452
273105448298078496512698878896456976090663420447798902191443793283001
949357096392170390483327088259620130177372720271862591991442827543742
```

(continued on next slide)

Code for a Universal Turing Machine

235135567513408422229988937441053430547104436869587640517812801943753
081387063994277282315642528923751456544389905278079324114482614235728
619311833261065612275553181020751108533763380603108236167504563585216
421486954234718742643754442879006248582709124042207653875426445413345
174856629157429990950262300973373813772416217274772361020678685400289
356608569682262014198248621698902609130940298570600174300670086896759
034473417412787425581201549366393899690581773859165405535670409282133
222163141097871081459978669599704509681841906299443656015145490488092
208448003482249207730403043188429899393135266882349662101947161910701
461968523192847482034495897709553561107027581748733327296678998798473
284098190764851272631001740166787363477605857245036964434897992034489
997455662402937487668839751404451665707750060513883991668814072545544
665222050724262392379211525318162512536305093172863142200406457130527
5802307665183351995689139748137504926429605010013651980186945639498

Turing's Universal Turing Machine

- Based on M -functions (subroutines with parameters) (Sicard 1997; Copeland 2004).
- The machine is composed by 12 symbols and 4.000 instructions, approximately (Sicard Ramírez 1998).

Small Universal Turing Machines

Notation

Let $\text{UTM}(m, n)$ be the class of universal Turing machines with m states and n symbols.

Small Universal Turing Machines

Notation

Let $\text{UTM}(m, n)$ be the class of universal Turing machines with m states and n symbols.

Theorem

If $\text{UTM}(m, n) \neq \emptyset$ then (Shannon 1956):

- (i) $\text{UTM}(2, n') \neq \emptyset$, where n' is at most $4mn + n$ and
- (ii) $\text{UTM}(m', 2) \neq \emptyset$, where $m' = (2^l - 1)m$ and l is the smaller integer such that $m \leq 2^l$.

Small Universal Turing Machines

Theorem

There exists universal Turing machines in the following classes (Rogozhin 1996; Neary and Woods 2012):

UTM(m, n)	Author(s)
(24, 2)	Rogozhin (1996)
(19, 2)	Baiocchi (2001)
(18, 2)	Neary and Woods (2007)
(15, 2)	Neary and Woods (2009)

(continued on next slide)

Small Universal Turing Machines

Theorem (continuation)

UTM(m, n)	Author(s)
(10, 3)	Rogozhin (1996)
(9, 3)	Neary and Woods (2009)
(7, 4)	Rogozhin (1996)
(6, 4)	Neary and Woods (2009)
(5, 5)	Rogozhin (1996)
(4, 6)	Rogozhin (1996)
(3, 10)	Rogozhin (1996)
(2, 18)	Rogozhin (1996)

Small Universal Turing Machines

Theorem

The following classes are empty (Rogozhin 1996; Neary and Woods 2012):

UTM(m, n)	Author(s)
$(m, 1)$	trivial
$(3, 2)$	Rogozhin (1996)
$(2, 3)$	Rogozhin (1996)
$(2, 2)$	Rogozhin (1996)
$(1, n)$	Herman (1968)

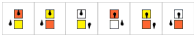
Wolfram Turing Machine

Announced May 14th, 2007: 5th Anniversary of the Publication of *A New Kind of Science*

THE WOLFRAM 2,3 TURING MACHINE RESEARCH PRIZE

\$25,000 prize

Is this Turing machine universal, or not?



The machine has 2 states and 3 colors, and is 596440 in Wolfram's numbering scheme. If it is universal then it is the smallest universal Turing machine that exists.

[BACKGROUND »](#) [TECHNICAL DETAILS »](#) [GALLERY »](#) [NEWS »](#)
[PRIZE COMMITTEE »](#) [RULES & GUIDELINES »](#) [FAQs »](#)

A universal Turing machine is powerful enough to emulate any standard computer. The question is: how simple can the rules for a universal Turing machine be?

*Since the 1960s it has been known that there is a universal 7,4 machine. In *A New Kind of Science*, Stephen Wolfram found a universal 2,5 machine, and suggested that the particular 2,3 machine that is the subject of this prize might be universal.*

The prize is for determining whether or not the 2,3 machine is in fact universal.

[What is a Turing Machine? »](#) | [Notable Universal Turing Machines »](#)

[INTERACTIVE DEMONSTRATIONS »](#) [PRINTABLE POSTER »](#)
[NKS|ONLINE »](#)

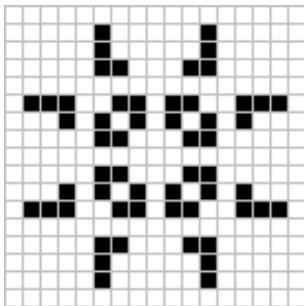
SPONSORED BY WOLFRAM RESEARCH & STEPHEN WOLFRAM



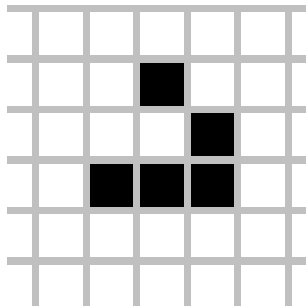
Conway's Game of Life

Examples

(From Wikipedia)



Pulsar (Oscillator)
Author: Jokey Smurf



Glider (Spaceship)
Author: Rodrigo Silveira Camargo

Conway's Game of Life

Rules

- (i) Any live cell with fewer than two live neighbours dies, as if caused by under-population.

Conway's Game of Life

Rules

- (i) Any live cell with fewer than two live neighbours dies, as if caused by under-population.
- (ii) Any live cell with two or three live neighbours lives on to the next generation.

Conway's Game of Life

Rules

- (i) Any live cell with fewer than two live neighbours dies, as if caused by under-population.
- (ii) Any live cell with two or three live neighbours lives on to the next generation.
- (iii) Any live cell with more than three live neighbours dies, as if by overcrowding.

Conway's Game of Life

Rules

- (i) Any live cell with fewer than two live neighbours dies, as if caused by under-population.
- (ii) Any live cell with two or three live neighbours lives on to the next generation.
- (iii) Any live cell with more than three live neighbours dies, as if by overcrowding.
- (iv) Any dead cell with exactly three live neighbours becomes a live cell, as if by reproduction.

Conway's Game of Life

Rules

- (i) Any live cell with fewer than two live neighbours dies, as if caused by under-population.
- (ii) Any live cell with two or three live neighbours lives on to the next generation.
- (iii) Any live cell with more than three live neighbours dies, as if by overcrowding.
- (iv) Any dead cell with exactly three live neighbours becomes a live cell, as if by reproduction.

Theorem

It is possible to codify a universal Turing machine in Conway's Game of Life (Rendell 2011).

The Halting Problem

The language of the halting problem

Let $\Sigma = \{0, 1\}$. The original Turing machine accepted by halting, no by final state.

$$H(M) := \{ w \in \Sigma^* \mid M \text{ halts given the input } w \}.$$

The Halting Problem

The language of the halting problem

Let $\Sigma = \{0, 1\}$. The original Turing machine accepted by halting, no by final state.

$$H(M) := \{ w \in \Sigma^* \mid M \text{ halts given the input } w \}.$$

We define the language of the halting problem by

$$L_{\text{hp}} := \{ \overrightarrow{(M, w)} \in \Sigma^* \mid w \in H(M) \}.$$

The Halting Problem

The language of the halting problem

Let $\Sigma = \{0, 1\}$. The original Turing machine accepted by halting, no by final state.

$$H(M) := \{ w \in \Sigma^* \mid M \text{ halts given the input } w \}.$$

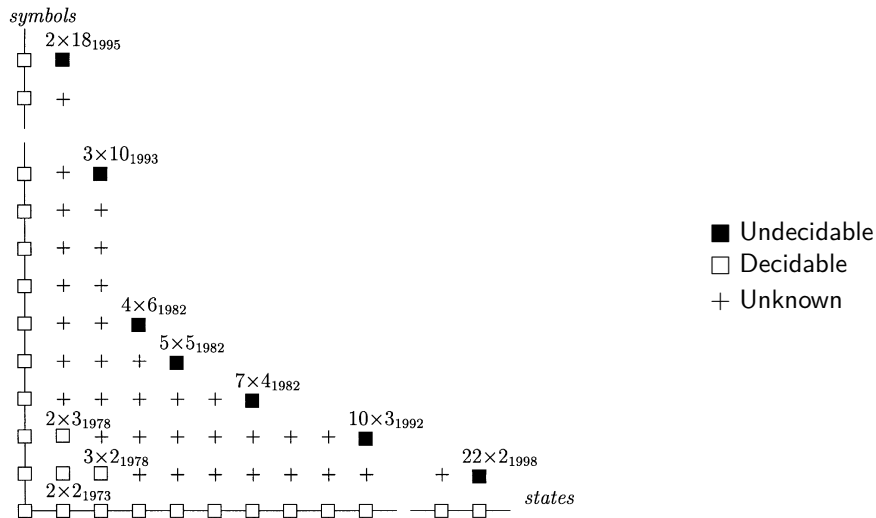
We define the language of the halting problem by

$$L_{\text{hp}} := \{ \overrightarrow{(M, w)} \in \Sigma^* \mid w \in H(M) \}.$$

Exercise 9.2.1

Show that L_{hp} is recursively enumerable but not recursive.

The Halting Problem: State of Art[†]



†Figure from (Margenstern 2000).

The Halting Problem

Remark

Although Turing neither introduced nor named the problem, he *“provided all the core ideas leading to a proof of the undecidability of the halting problem”*

(Hamkins and Nenu 2026, p. 15).

The Halting Problem

Remark

Although Turing neither introduced nor named the problem, he *“provided all the core ideas leading to a proof of the undecidability of the halting problem”*

(Hamkins and Nenu 2026, p. 15).[†]

[†]The halting problem was introduced by (Kleene 1974, p. 382) and it was named by Davis (1958, p. 70). See (Lucas 2021; Hamkins and Nenu 2026).

The Halting Problem

Practical approach

*“In contrast to popular belief, proving termination **is not always impossible**.”*

(Cook, Podelski and Rybalchenko 2011, p. 1)

References

- Claudio Baiocchi (2001). Three Small Universal Turing Machines. In: Machines, Computations, and Universality (MCU 2001). Ed. by Maurice Margenstern and Yurii Rogozhin. Vol. 2055. Lecture Notes in Computer Science. Springer, pp. 1–10. DOI: [10.1007/3-540-45132-3_1](https://doi.org/10.1007/3-540-45132-3_1) (cit. on p. 20).
- Byron Cook, Andreas Podelski and Andrey Rybalchenko (2011). Proving Program Termination. Communications of the ACM 54.5, pp. 88–98. DOI: [10.1145/1941487.1941509](https://doi.org/10.1145/1941487.1941509) (cit. on p. 36).
- Martin Davis (1958). Computability and Unsolvability. McGraw-Hill (cit. on p. 35).
- Joel David Hamkins and Theodor Nenu (2026). Did Turing Prove the Undecidability of the Halting Problem? Journal of Logic and Computation 36.1, exaf075. DOI: [10.1093/logcom/exaf075](https://doi.org/10.1093/logcom/exaf075) (cit. on pp. 34, 35).
- Gabor T. Herman (1968). The Uniform Halting Problem for Generalized One State Turing Machines. In: 9th Annual Symposium on Switching and Automata Theory (SWAT 1968). IEEE, pp. 368–372. DOI: [10.1109/SWAT.1968.36](https://doi.org/10.1109/SWAT.1968.36) (cit. on p. 22).
- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on pp. 2, 9).
- Stephen Cole Kleene [1952] (1974). Introduction to Metamathematics. Seventh reprint. Bibliotheca Mathematica: A Series of Monographs on Pure and Applied Mathematics. Volume 1. North-Holland (cit. on p. 35).
- Salvador Lucas (2021). The Origins of the Halting Problem. Journal of Logical and Algebraic Methods in Programming 121, p. 100687. DOI: [10.1016/j.jlamp.2021.100687](https://doi.org/10.1016/j.jlamp.2021.100687) (cit. on p. 35).

References

- Maurice Margenstern (2000). Frontier Between Decidability and Undecidability: A Survey. *Theoretical Computer Science* 231.2, pp. 217–251. DOI: [10.1016/S0304-3975\(99\)00102-4](#) (cit. on p. 33).
- Turlough Neary and Damien Woods (2007). Four Small Turing Machines. In: *Machines, Computations, and Universality (MCU 2007)*. Ed. by Jérôme Durand-Lose and Maurice Margenstern. Vol. 4664. *Lecture Notes in Computer Science*. Springer, pp. 242–254. DOI: [10.1007/978-3-540-74593-8_21](#) (cit. on p. 20).
- (2009). Four Small Turing Machines. *Fundamenta Informaticae* 91.1, pp. 123–144. DOI: [10.3233/FI-2009-0036](#) (cit. on pp. 20, 21).
- (2012). The Complexity of Small Universal Turing Machines: A Survey. In: *Theory and Practice of Computer Science (SOFSEM 2012)*. Ed. by Mária Bielíková, Gerhard Friedrich, Georg Gottlob, Stefan Katzenbeisser and György Turán. Vol. 7147. *Lecture Notes in Computer Science*. Springer, pp. 385–405. DOI: [10.1007/978-3-642-27660-6_32](#) (cit. on pp. 20, 22).
- Roger Penrose (1991). *The Emperor's New Mind*. Penguin Books (cit. on p. 15).
- Paul Rendell (2011). A Universal Turing Machine in Conway's Game of Life. In: *Proceedings of the 2011 International Conference on High Performance Computing & Simulation (HPCS 2011)*. Ed. by Waleed W. Smari. IEEE, pp. 764–772. DOI: [10.1109/HPCSim.2011.5999906](#) (cit. on pp. 25–29).

References

- Yurii Rogozhin (1996). Small Universal Turing Machines. Theoretical Computer Science 168, pp. 215–240. DOI: [10.1016/S0304-3975\(96\)00077-1](https://doi.org/10.1016/S0304-3975(96)00077-1) (cit. on pp. 20–22).
- Claude E. Shannon (1956). A Universal Turing Machine with Two Internal States. In: Automata Studies. Ed. by C. E. Shannon and J. McCarthy. Vol. 34. Annals of Mathematics Studies. Princeton University Press, pp. 157–165 (cit. on pp. 18, 19).
- Andrés Sicard (1997). Máquina Universal de Turing: Algunas Indicaciones para su Construcción. Revista Universidad EAFIT 33.108, pp. 61–106 (cit. on p. 17).
- Andrés Sicard and B. Jack Copeland (2004). Appendix: Subroutines and M-functions. In: The Essential Turing. Vol. 317. 1–3, pp. 54–57. DOI: [10.1016/j.tcs.2003.12.014](https://doi.org/10.1016/j.tcs.2003.12.014) (cit. on p. 17).
- Andrés Sicard Ramírez (1998). Máquinas de Turing Dinámicas: Historia y Desarrollo de una Idea. MA thesis. Departamento de Informática y Sistemas. Universidad EAFIT (cit. on p. 17).